INCH POUND

MIL-C-915/82A
29 September 1989
SUPERSEDING
MIL-C-915/82
18 June 1985

MILITARY SPECIFICATION SHEET

# CABLE, POWER ELECTRICAL, ONE TWISTED PAIR NO. 16 GAUGE FLUOROPOLYMER INSULATION, FOR SUBMARINE OUTBOARD USE ONLY (NOT FOR INBOARD USE), TYPE 1PR-16

This specification sheet is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation: MIL-C-915.

REQUIREMENTS:

Qualification required.

## Construction

First - Copper conductor, tin coated, AWG size 16, class B stranding.

Second - Insulation of ethylene tetrafluoroethylene (ETFE). Nominal wall thickness 0.010 inch. Maximum overall diameter 0.082 inch.

Third - Two insulated conductors, one black and one white, twisted together with a lay of 1-1/4 ± 1/8 inch.

Fourth - Cable jacket of black polyether polyurethane applied so as to completely fill the pair valleys. The cable jacket material shall B.F. Goodrich Co. ESTANE 58300 or 58853; or Mobay Chemical Co. TEXIN 985A. Nominal wall thickness 0.105 inch. Overall diameter to be 0.380 to 0.400 inch.

AMSC N/A FSC 6145

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EXAMINATION AND TESTS:

| | Requirements |
|---|---|
| Basic electrical: | |
| Conductor resistance - at 20°C, ohms/1000 feet, maximum | 4.24 |
| Voltage withstand - volts, root mean square, minimum | |
| Conductor to conductor | 2000 |
| Conductor to water | 2000 |
| Insulation dielectric (on 100 percent of insulated conductor) - test in accordance with MIL-W-81381, using a test voltage of 8.0 kV (peak), minimum. | |
| Insulation resistance - megohms/1000 feet, minimum | 5000 |
| Group A: | |
| Visual and dimensional | No failure |
| Group B: | |
| Cold bending, cable - at minus 40 ± 2°C over a 3-inch diameter mandrel | |
| Physicals (unaged) | |
| Insulation | |
| Tensile strength - lbf/in², minimum | 5000 |
| Elongation - percent, minimum | 100 |
| Cable jacket | |
| Tensile strength - lbf/in², minimum | 4000 |
| Elongation - percent, minimum | 500 |
| Group C: | |
| Physicals (aged) | |
| Insulation | |
| Air oven (168 hours at 180°C) | |
| Tensile strength - percent of unaged, minimum | 80 |
| Elongation - percent of unaged, minimum | 90 |
| Cable jacket | |
| Air oven (168 hours at 80°C) | |
| This test shall be in accordance with MIL-C-915 | No failure |
| Hot oil immersion | |
| Tensile strength - percent of unaged, minimum | 60 |
| Elongation - percent of unaged, minimum | 80 |

QUALIFICATION INSPECTION:

Qualification inspection shall include basic electrical, plus all of groups A, B and C.



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UNIT ORDERING LENGTH: 500 feet (minimum).

NOTE: Not for Air Force use.

Revision letters are not used to denote changes due to the extensiveness of the changes.

Custodians:
Army - MI
Navy - SH

Preparing activity:
Navy - SH
(Project 6145-1127-19)

Review activities:
Army - CR, ER
Navy - EC
DLA - IS

User activities:
Army - AL, AR, ME
Navy - CG



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